

hydro-tl-ews

Transfer Learning for Hydrological Early Warning in Data-Scarce Regions

Project README — inputs, how to run, and expected outputs

An Entity-Aware LSTM (EA-LSTM) is pre-trained on CAMELS-US donor catchments, transferred to a data-scarce target basin (~2-year warmup), and evaluated under a rolling-origin **walk-forward** protocol. Outputs include continuous streamflow skill (NSE / KGE / PBIAS), flood/drought early-warning probabilities, optional SHAP attributions, and a 7-basin multi-regime study.

Primary single-target default: **Merced River at Happy Isles** (11264500).

Expected inputs

Input	Required for	Notes
Python ≥ 3.10 + deps	Always	See Setup
CAMELS-US under data/	All real stages	~14 GB; see data/README.md
YAML config under configs/	Every stage	Points at data paths, periods, checkpoints
Upstream checkpoint	Fine-tune / zero-shot / walk-forward / multi-target	e.g. results/checkpoints/pret_rain_subset200.pt

No CAMELS is needed for the **smoke** path (synthetic basins).

CAMELS layout (required)

```
data/
basin_dataset_public_v1p2/
basin_mean_forcing/daymet/...
usgs_streamflow/...
camels_attributes_v2.0/
camels_topo.txt
camels_clim.txt
...
```

Upstream citation notes for the attributes package live in `data/readme.txt`.

Setup

```
cd hydro-tl-ews
python3 -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
pip install -e . # optional; enables `hydro-tl-ews` CLI
```

Conda alternatives: `environment.yml` (GPU-oriented) or `environment-cpu.yml`.

Verify CAMELS (after download):

```
python -c "
from pathlib import Path
root = Path('data')
assert (root / 'basin_dataset_public_v1p2').is_dir()
assert (root / 'camels_attributes_v2.0' / 'camels_topo.txt').is_file()
print('CAMELS layout OK')
"
```

How to run

All commands assume the repo root and an activated environment.

If you did not `pip install -e .`, prefix with:

```
export PYTHONPATH="$PWD/src${PYTHONPATH:+:$PYTHONPATH}"
```

1. Smoke test (no CAMELS, minutes)

```
python scripts/run_experiment.py --config configs/smoke_test.yaml
# or explicitly:
python scripts/run_experiment.py --config configs/smoke_test.yaml --smoke
python -m pytest -q
```

Writes under `results/smoke/` (synthetic end-to-end check).

1b. Mini sample CAMELS (real loader, a few minutes)

No full CAMELS download required. Generates a tiny CAMELS-compatible archive

and runs pretrain → fine-tune → walk-forward through CamelsDataset:

```
python scripts/make_sample_camels.py # -> data/sample_camels/ (~4 MB)
python scripts/run_sample_pipeline.py # -> results/sample/
```

2. Single stage (config-driven)

```
python scripts/run_experiment.py --config configs/<stage>.yaml
# after editable install:
hydro-tl-ews --config configs/<stage>.yaml
```

Stage config	Role	Needs
pretrain_subset200.yaml	~200 similar donors, 30 epochs (default laptop/GPU path)	CAMELS
pretrain.yaml	Full CAMELS regional model (heavy; multi-day on laptop)	CAMELS
zero_shot.yaml	Pretrained model, no fine-tune	pretrain_subset200.pt
finetune_conservative.yaml	Approach A (head-only)	pretrain_subset200.pt
finetune_progressive.yaml	Approach B (progressive unfreeze)	pretrain_subset200.pt
local_baseline.yaml	Train from scratch on warmup	CAMELS
walk_forward.yaml	Operational 2011–2014 eval + EWS (+ SHAP)	finetune_conservative.pt
walk_forward_progressive.yaml	Same with Approach B refits	finetune_progressive.pt
min_data_sensitivity.yaml	Warmup length sweep	pretrain_subset200.pt

3. Full single-target pipeline

```
bash scripts/run_full_pipeline.sh
```

Order: pretrain_subset200 → zero_shot → finetunes → local_baseline →

walk_forward → min_data_sensitivity. Log: results/logs/full_pipeline.log.

For **full CAMELS** pretrain (feeds multi-target):

```
python scripts/run_experiment.py --config configs/pretrain.yaml
# -> results/checkpoints/pretrain.pt
```

4. Seven-basin multi-target study

Requires `results/checkpoints/pretrain.pt` (from `configs/pretrain.yaml`).

```
python scripts/run_multi_target.py # all basins, resumable
python scripts/run_multi_target.py 11264500 # one basin
python scripts/run_multi_target.py --summary
```

Outputs under `results/multi_target/<gauge_id>/` and `results/multi_target/summary.csv`.

5. Figures / manuscript (optional)

Need completed `results/` artifacts and extra deps (`reportlab`, `python-docx` for paper builders — not in `core requirements.txt`).

```
python scripts/figures/make_figures.py
python scripts/paper/build_paper.py
python scripts/paper/build_manuscript_docx.py
```

Expected outputs

Everything lands under `results/` (gitignored).

Artifact	Typical path	Contents
Checkpoint	<code>results/checkpoints/*.pt</code>	Model weights + config dict
Train history	<code>results/history/*.json</code>	Epoch train/val losses
Zero-shot metrics	<code>results/zero_shot_metrics.json</code>	NSE, KGE, PBIAS, <code>n_samples</code>
Zero-shot series	<code>results/zero_shot_predictions.csv</code>	Daily obs / pred
Walk-forward series	<code>results/walk_forward.parquet</code>	date, observed, predicted, bias
Walk-forward metrics	<code>results/walk_forward_metrics.json</code>	Continuous + EWS AUC/F1/Brier

Artifact	Typical path	Contents
Warning table	results/walk_forward_warnings.csv	Labels + probs by lead time
SHAP	results/shap_global_importance.csv	Mean SHAP by feature
Min-data table	results/min_data_sensitivity.csv	Skill vs warmup months
Smoke	results/smoke/	Synthetic summary + CSV
Multi-target	results/multi_target/...	Per-basin copies of the above

Device selection is automatic: **CUDA** → **MPS (Apple Silicon)** → **CPU**

(hydro_tl_ews.utils.device.training_device).

Repository layout

```
hydro-tl-ews/
+-- src/hydro_tl_ews/ # importable library (data, models, training, eval, xai)
+-- scripts/ # CLI + stage runners + analysis / figures / paper
+-- configs/ # YAML experiments (+ configs/multi_target/)
+-- tests/ # pytest
+-- data/ # CAMELS (not versioned) — see data/README.md
+-- figures/ # checked-in paper figures
+-- docs/ # running guide, outputs, known limitations
+-- results/ # generated (gitignored)
```

Method summary

- **Model** — EA-LSTM (Kratzert et al., 2019): static attributes drive a

time-invariant input gate; dynamic forcings drive forget / candidate / output.

- **Transfer** — regional pre-train → target fine-tune. Approach A: head-only.

Approach B: progressive unfreeze of the last LSTM parameter group at a smaller LR.

- **Evaluation** — 2-year warmup, 90-day walk-forward refits, online bias

correction, validation tail + best-weight restore, refit_train_start to keep

the scarce-data protocol honest.

- **Early warning** — at-site Q5/Q95/Q99 (fit on pre-eval years); Gaussian residual mapping; scored with AUC, F1, Brier (and BSS vs DOY climatology in analysis scripts).
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Documentation

- [docs/README.pdf](#) — downloadable PDF of this README
- [docs/RUNNING.md](#) — detailed run recipes and compute notes
- [docs/OUTPUTS.md](#) — file formats and how to read them
- [docs/KNOWN_LIMITATIONS.md](#) — remaining science/UX caveats
- [data/README.md](#) — CAMELS download and layout

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